

Esports Data Analytics Report (Tableau Project)

1. Problem Statement

The objective of this project is to analyze the global gaming and esports industry using interactive dashboards. The analysis focuses on understanding industry growth in terms of active players, revenue, esports tournaments, prize pools, viewers, gaming platforms, and player behavior.

The goal of this project is to transform complex gaming data into meaningful visual insights that help identify industry trends and growth patterns.

2. Dataset Overview

The dataset used in this project contains information related to the global gaming and esports ecosystem between **2010 and 2025**.

The dataset includes the following key fields:

Gaming Industry Metrics

- Active Players
- Gaming Revenue
- Esports Revenue
- Esports Viewers

Esports Metrics

- Total Tournaments
- Professional Players
- Prize Pool

Player Behavior Metrics

- Female Gamer Percentage
- Mobile Gaming Share
- Internet Penetration
- Average Spending per Gamer

Geographic Data

- Region
- Country

Gaming Market Data

- Gaming Platform (PC, Console, Mobile)
- Gaming Genre (FPS, RPG, MOBA, Sports, Strategy)

This dataset helps analyze different aspects of the gaming industry including financial growth, player distribution, market trends, and esports performance.

3. Data Preprocessing

Before creating the dashboards, the dataset was cleaned and prepared using the following preprocessing steps:

- Checked and removed duplicate records.
- Verified the accuracy of numeric fields such as revenue and prize pool.
- Checked for missing or null values.
- Standardized column names for better readability.
- Verified correct data types for each column.
- Prepared the dataset for visualization in Tableau.

These preprocessing steps ensured that the dataset was clean, consistent, and ready for analysis.

4. Dashboards Created

Three dashboards were created in Tableau to analyze the gaming and esports ecosystem from different perspectives.

4.1 Dashboard 1 – Global Gaming Overview

Purpose:

To provide a high-level overview of the global gaming industry between 2010 and 2025.

Key KPIs:

- Total Active Players
- Total Viewers
- Total Gaming Revenue
- Total Esports Revenue

Visualizations included:

- Gaming Revenue Trend
- Esports Revenue Trend

- Gaming Revenue by Country (Map)
- Top Gaming Genre
- Active Players by Region

This dashboard helps understand the overall growth of the gaming industry and identify regions with high gaming participation.

4.2 Dashboard 2 – Esports Analytics

Purpose:

To analyze esports tournaments, professional players, and prize pools.

Key KPIs:

- Total Tournaments
- Total Professional Players
- Total Prize Pool
- Total Esports Viewers

Visualizations included:

- Prize Pool Trend
- Pro Players by Region
- Esports Viewers Growth
- Prize Pool by Country
- Tournaments by Country

This dashboard provides insights into the competitive esports ecosystem and highlights the growth of esports tournaments and audience engagement.

4.3 Dashboard 3 – Player Behavior and Technology

Purpose:

To analyze gamer behavior, gaming platforms, and technology trends.

Key KPIs:

- Internet Penetration
- Mobile Gaming Share
- Female Gamer Percentage
- Average Spending per Gamer

Visualizations included:

- Female Gamers by Region
- Average Spending by Country
- Genre Distribution
- Mobile Gaming Trend
- Platform Popularity

This dashboard helps understand gaming habits, platform usage, and player demographics.

5. Key Insights

From the analysis of the dashboards, several important insights were identified:

- The global gaming industry has experienced significant growth in revenue from 2010 to 2025.
- Esports revenue and prize pools have increased steadily over time.
- Some regions have a significantly higher number of active gamers.
- Mobile gaming has become one of the most popular gaming platforms.
- Certain gaming genres such as RPG and FPS dominate the gaming market.
- Esports tournaments and viewership have increased significantly in recent years.

These insights highlight the rapid expansion and popularity of the global gaming ecosystem.

6. Recommendations

Based on the analysis, the following recommendations can be made:

- Invest more in regions with high gaming participation.
 - Focus on mobile gaming platforms as they have a large user base.
 - Support esports tournaments to increase audience engagement.
 - Promote popular gaming genres to attract more players.
 - Improve marketing strategies based on player behavior and regional trends.
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7. Conclusion

The Esports Data Analytics project provides valuable insights into the global gaming industry. Using Tableau dashboards, the project visualizes trends related to revenue, players, tournaments, and gaming platforms.

The dashboards help transform raw gaming data into meaningful insights that highlight the growth and future potential of the esports ecosystem. These visualizations can support better decision-making for gaming companies and industry stakeholders.